

Fig. 5: Actual-size PCB of the transmitter unit

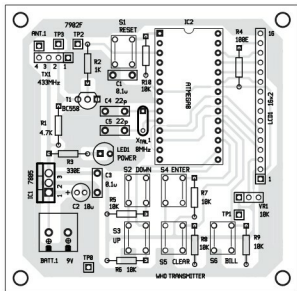


Fig. 6: Component layout of the PCB of the transmitter unit

if CKOPT fuse bit is programmed. Capacitor values for C4 and C5 in transmitter and C9 and C10 in receiver should always be equal for both crystals and resonators. The optimal value of capacitors depends on the crystal or resonator in use, amount of stray capacitance and electromagnetic noise of the environment.

Reset inputs. ATmega8 has four sources of reset, namely, power-on reset, external reset, watchdog reset and brown-out reset. Switch S1 is used as an external reset to reset MCU IC2, when a low voltage level

is present on the reset pin for longer than the minimum pulse length as defined in the data-sheet. Similarly, switch S7 is used as an external reset for IC4.

The transmitter unit will be energised as soon as the power supply is applied and the MCU will be reset due to the action of power-on reset circuit formed by capacitor C1 and resistor R10. The MCU will initially be in idle state and the LCD back light will be in low brightness in order to save power.

Welcome (in Malayalam) message language will be displayed in the idle state. In this state, the system checks the status of switches S2 through S6 connected to port B of ATmega8. When the user presses any switch, the system breaks from the idle state and jumps to the menu directly.

Now, the program is in continuous main loop. Since the LCD display contains only two lines, it can display only two messages at a time. The current selected item is highlighted by enclosing it within > and < symbols. The LCD display is updated only when a key is pressed, thereby avoiding updates in the loop.

By pressing Up or Down switch, the user can go to Previous or Next menu. The quantity of selected item can also be set using Up and Down. The price of the selected item is also displayed in the second line of the display. If the user wishes to cancel the order, he can press Clear, and

"Cancelled.." will be displayed on the LCD and the program will jump back to the main menu.

Pressing Enter will confirm the order, and "Sending.." will be displayed on the LCD and a code will be generated internally. This code will contain the item code and the number of items ordered. This code is transmitted serially using USART communication technique in AVR. After sending, "Sent" is displayed on the LCD and the program jumps to the main menu.

By pressing Up or Down, the user can review the order, including items ordered and the quantity. After ordering the required items, if the user wishes to view and verify the bill, he or she can press Bill and it will show the total amount in rupees.

Game. While the user waits for food, he or she can play Shooter game by pressing Game. The game features three custom characters and Enter shoots enemies.

Software program

Programming of the AVR is done using embedded C language. It is similar to C language but includes all functionalities of C as well as access to AVR pins, peripherals and controls. The Cross compiler used here is AVR-gcc, which is an open source compiler.

Hex codes generated are burnt into the MCUs for transmitter and receiver units. Working of the program is explained as comments in transmitter and receiver source codes.

The program jumps to the main function where the object code actually starts. Enter key causes the program to jump to the sub-loop of ordering. Pressing Enter will confirm the order and the entered quantity is stored in the item quantity variable in the program. The key de-bouncing problem is eliminated by the software by adding some delays for every key pressed.

Construction and testing

An actual-size, single-side PCB of the transmitter unit is shown in Fig. 5 and its component layout in Fig. 6. An actual-size, single-side PCB